

# When Models Commit: Candidate Formation and Winner Isolation Across Model Depth

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## Abstract

1        Language models produce one final answer, but the forward pass may in-  
 2        involve two stages before that answer appears: the model may first narrow  
 3        the set of answers that still look possible, and only later commit to one  
 4        winner. We study that gap in multiple-choice question answering.

5        For each prompt, we end the rendered question with the literal suffix "Fi-  
 6        nal answer: " and score exact continuations layer by layer. On the primary  
 7        path, the continuation is the option text itself. This lets us track when  
 8        the eventual answer first enters a small plausible-answer set, when it be-  
 9        comes the stable top prediction, and when it opens a persistent margin over  
 10       the runner-up. We run the analysis on nine instruction-tuned open-weight  
 11       models from the Qwen, Gemma, and Llama families on a 1,000-example  
 12       slice of ARC-Challenge, CommonsenseQA, and the full 11-question MMLU  
 13       abstract-algebra validation slice.

14       The main pattern is clear. In the pooled primary analysis, by 20% of  
 15       normalized depth the eventual answer is already in the plausible set for  
 16       72% of evaluable cases, but stable commitment has happened in only 4%.  
 17       All 18 evaluable ARC-Challenge and CommonsenseQA model-dataset cells  
 18       are positive under the preregistered primary rule. The 3 evaluable MMLU  
 19       cells are also positive, but that benchmark slice is mostly underpowered  
 20       because it contains only 11 questions.

21       We also score two answer-format controls after the same prompt suffix:  
 22       the literal continuation "The answer is {option}" and the answer letter  
 23       alone. These controls often change both which cases are evaluable and  
 24       which answer wins at the final layer, so they are useful checks on answer-  
 25       format effects rather than replacements for exact option-text scoring. In  
 26       the setting we study, models usually treat the eventual answer as a serious  
 27       possibility well before they fully commit to it.

## 28    1 Introduction

29    A multiple-choice answer hides more than one decision. A model can first narrow the field  
 30    to a few answers that still seem possible and only later settle on one final winner. Standard  
 31    evaluation only shows the last step. It tells us what answer came out, but not when that  
 32    answer first became a serious option inside the network.

33    This paper asks a simple question. Does the eventual answer become one of the plausible  
 34    answers before the model writes the margin that finally separates it from the runner-up?  
 35    Figure 1 previews the answer. In the pooled primary analysis, by 20% of normalized depth  
 36    the eventual answer is already in the plausible answer set for 72% of evaluable cases, but  
 37    only 4% are already stably committed.

38    We answer that question in a very literal way. We build a normal multiple-choice prompt,  
 39    end it with the suffix "Final answer: ", and then score the exact strings the model could write  
 40    next. On the main path, that string is the full option text itself, such as "photosynthesis".  
 41    So instead of asking an external probe to guess the answer, we watch how the model's own  
 42    answer strings rise and fall across depth.

| Aspect        | Configuration                                                                                                                                                                                                                                          |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Models        | Nine instruction-tuned checkpoints from Qwen 2.5 Instruct (1.5B, 3B, 7B), Gemma 2 IT (2B, 9B, 27B), and Llama Instruct (3.2 1B, 3.2 3B, 3.1 8B).                                                                                                       |
| Dataset slice | ARC-Challenge (195), CommonsenseQA (794), and MMLU Abstract Algebra (11).                                                                                                                                                                              |
| Prompting     | Model-specific chat or raw prompt templates. Every prompt ends with the literal suffix "Final answer:".                                                                                                                                                |
| Readouts      | Primary: the exact option text after "Final answer:", for example "photosynthesis". Controls: the literal continuation "The answer is {option_text}" and the bare answer letter, such as "A".                                                          |
| Permutations  | Identity ordering and reverse ordering, kept paired on the same example subset.                                                                                                                                                                        |
| Primary rule  | For each model-dataset cell, report the median commitment lag $\Delta_{\text{margin}} = d_{\text{margin}} - d_{\text{id}}$ with a grouped-bootstrap 95% confidence interval; the cell is positive iff the lower confidence bound is greater than zero. |

Table 1: Reader-facing summary of the experiment design. The main question stays fixed throughout the paper: scoring the exact option text is primary, the two alternative continuation strings are controls, and identity versus reverse answer order is kept paired on the same example subset.

Our setting is intentionally narrow and easy to audit. We evaluate nine instruction-tuned open-weight models from the Qwen 2.5 Instruct, Gemma 2 IT, and Llama Instruct families on a 1,000-example slice spanning ARC-Challenge, CommonsenseQA, and MMLU abstract algebra. The claim of the paper is limited to that setting. Within it, the pattern is consistent: the eventual answer usually enters the plausible answer set well before it becomes stably isolated from the runner-up. All 18 evaluable ARC-Challenge and CommonsenseQA model-dataset cells are positive under the preregistered primary rule (Figure 2, Table 2). The MMLU abstract-algebra slice is much smaller, so it is too small to carry much weight on its own.

The paper makes three contributions. First, it gives a simple measurement framework for separating “still possible” from “finally chosen” without leaving the answer strings that appear in the task itself. Second, it shows a large and consistent gap between those two stages in the pooled curves and in every evaluable ARC-Challenge and CommonsenseQA cell. Third, it shows that small changes to answer format, such as switching from full option text to an answer letter, can change both coverage and winners. That makes those alternatives useful controls, but not replacements for exact option-text scoring.

## 2 Setup and measurement

Table 1 summarizes the experiment design. The model panel contains nine instruction-tuned checkpoints from three families: Qwen 2.5 Instruct (Qwen Team, 2024), Gemma 2 IT (Google, 2024), and Llama Instruct (AI@Meta, 2024). The dataset slice contains 195 ARC-Challenge prompts (Clark et al., 2018), 794 CommonsenseQA prompts (Talmor et al., 2019), and the full 11-question MMLU abstract-algebra validation slice (Hendrycks et al., 2021). The MMLU count is small because this benchmark slice itself is small in our setup; we did not drop a larger set of abstract-algebra questions.

For every model, we render a normal multiple-choice prompt and end it with the literal suffix "Final answer: ". After that suffix, we score three possible continuations:

- the exact option text itself, such as "photosynthesis";
- the literal template "The answer is {option\_text}";
- the bare answer letter, such as "A".

The exact option-text path is the preregistered primary analysis. The other two are controls. They keep the same prompt and the same answer choices, but they change what exact continuation string we score after "Final answer: ".

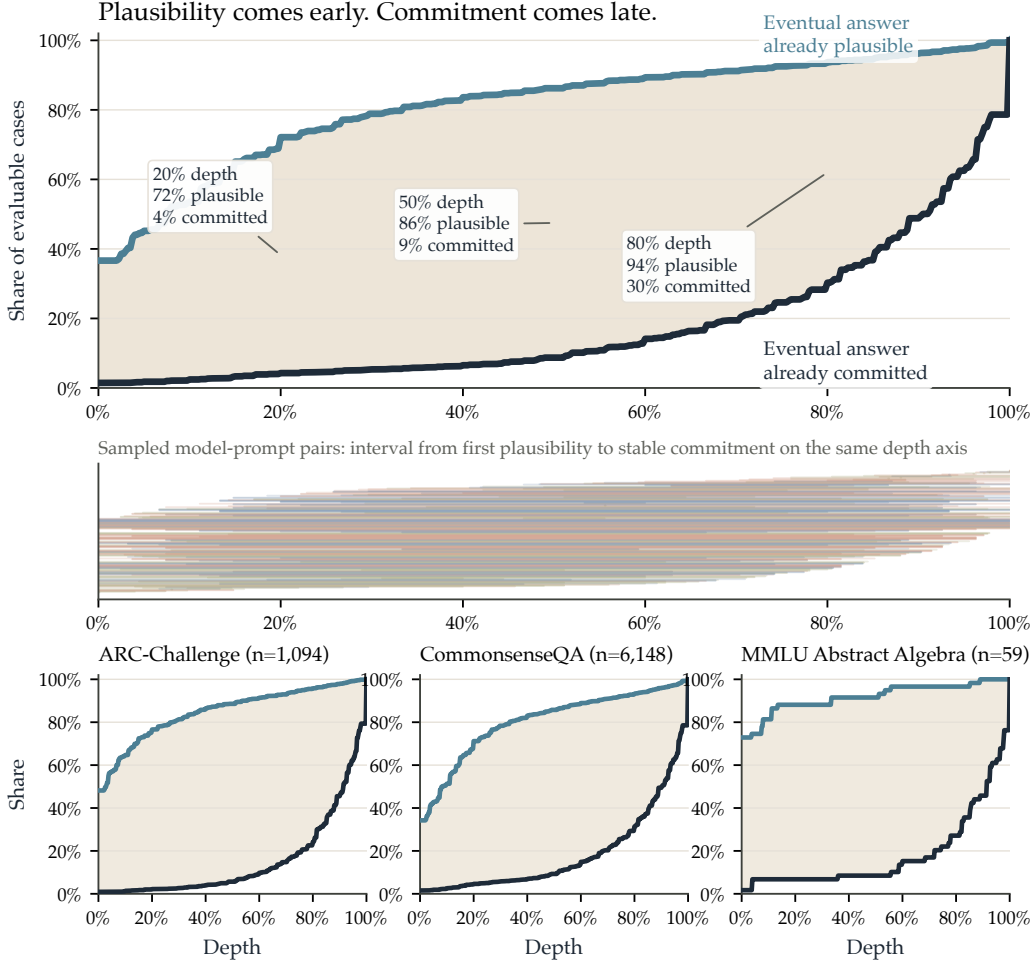


Figure 1: The eventual answer becomes plausible much earlier than it becomes committed. The top panel pools all evaluable model-prompt pairs from the primary analysis path (semantic\_exact with identity ordering). The light curve asks: by this depth, in how many cases is the eventual answer already one of the plausible answers? The dark curve asks: by this depth, in how many cases has the model already opened a stable winner-runner margin that it keeps through the end of the network? The band between the curves is the main object in the paper: cases where the answer already looks plausible but the model has not fully committed yet. The interval strip shows a sample of individual cases on the same normalized-depth axis, and the three bottom panels show the same pattern separately for ARC-Challenge, CommonsenseQA, and MMLU abstract algebra.

75 We score these continuations under teacher forcing, one layer at a time. In plain terms, we  
 76 do not let the model free-generate a long answer. We stop at "Final answer: " and ask  
 77 how much probability the network assigns to each allowed continuation at each layer. Let  
 78  $p_\ell(o)$  denote the normalized probability of option  $o$  at layer  $\ell$ . We sort options by  $p_\ell(o)$  and  
 79 define a plausible answer set  $C_\ell$  as the smallest prefix whose cumulative probability reaches  
 80  $\tau = 0.8$ . If the cutoff lands inside a numerical tie, we mark that layer as ambiguous instead  
 81 of forcing an arbitrary choice. In plain language,  $C_\ell$  is the small set of answers that the  
 82 model still treats as seriously possible at layer  $\ell$ .

Our primary anchor is the eventual answer, meaning the option that wins at the final layer under the primary readout. Denote it by  $o^*$ . We then track three depth events:

$$d_{\text{id}} = \min\{\ell : o^* \in C_\ell\}, \quad (1)$$

$$d_{\text{top1}} = \min\{\ell : o^* \text{ is the unique top-1 option at every later layer}\}, \quad (2)$$

$$d_{\text{margin}} = \min\{\ell : p_t(o^*) - p_t(o_{(2)}) \geq \delta \text{ for all } t \geq \ell\}, \quad (3)$$

where  $o_{(2)}$  is the runner-up option and  $\delta = 0.15$ . In words,  $d_{\text{id}}$  is when the eventual answer first enters the plausible set,  $d_{\text{top1}}$  is when it becomes the unique top prediction and stays there, and  $d_{\text{margin}}$  is when it opens a stable margin over the runner-up and never loses it.

The paper’s main depth-gap statistic is

$$\Delta_{\text{margin}} = d_{\text{margin}} - d_{\text{id}}. \quad (4)$$

This quantity asks a simple question: once the eventual answer is already in the plausible set, how many layers does the model still take before it clearly commits? A case is evaluable for the primary statistic when both  $d_{\text{id}}$  and  $d_{\text{margin}}$  are defined.

The preregistered primary rule is applied separately to each model-dataset cell on the primary path (semantic\_exact, identity ordering). For each cell we report the median of  $\Delta_{\text{margin}}$  over valid examples and estimate a grouped-bootstrap 95% confidence interval by resampling example IDs. A cell is positive if the observed median is greater than zero and the lower confidence bound is also greater than zero. We also report two additional summaries. The first is a robustness check based on the fraction of valid examples with positive lag. The second repeats the analysis while anchoring on the gold answer instead of the model’s final winner. These summaries help interpret the result, but they do not determine the paper’s main decision rule.

This setup intentionally differs from probe-based approaches such as the tuned lens (Belrose et al., 2023). We do not train a separate probe to guess the future answer from hidden states. Instead, we keep the task fixed and score the same answer strings the model is being asked to continue with.

### 3 Main result: early plausibility, late commitment

Figure 1 is the central result of the paper. The two cumulative curves are far apart. By 20% of normalized depth, the eventual answer is already plausible for 72% of evaluable cases, but it is already committed for only 4%. The gap stays large through the middle of the network. At 50% of normalized depth, 86% of evaluable cases have already admitted the eventual answer into the plausible set, while only 9% have already reached stable commitment. In raw layer units, the pooled grouped-bootstrap median commitment lag is 19.2 layers with a 95% confidence interval of [18.9, 19.5].

The dataset panels show that this is not coming from just one benchmark. ARC-Challenge and CommonsenseQA both show the same broad shape: plausibility rises early, while commitment stays much lower until later in depth. The MMLU abstract-algebra panel points in the same direction in the cases we can evaluate, but it is much noisier because the whole slice contains only 11 questions. So ARC-Challenge and CommonsenseQA carry most of the evidence, while MMLU is better read as a small supporting slice than as a decisive test on its own.

Figure 2 and Table 2 translate the pooled picture into the preregistered cell-level decision rule. All 18 evaluable ARC-Challenge and CommonsenseQA cells are positive, and there are no evaluated negatives. Across all three datasets, 21 cells are positive and 6 are insufficient. Within MMLU specifically, 3 cells are positive and 6 are insufficient. That is what we would expect from a very small benchmark slice under a minimum-valid-example rule.

The family-level summaries tell the same story. Qwen contributes eight positive primary cells, Gemma contributes seven, and Llama contributes six. In every family, the missing positives are insufficient MMLU cells rather than evaluated negatives. So the result is broad across the chosen instruct-model panel, even though the exact lag size still varies with model depth and dataset difficulty.

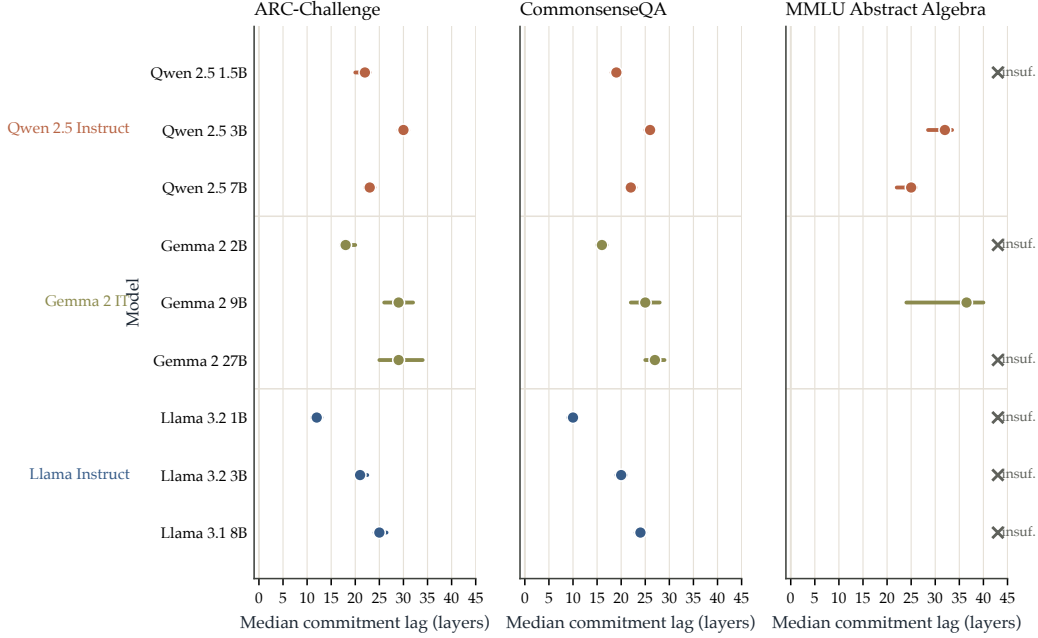


Figure 2: Every evaluable ARC-Challenge and CommonsenseQA cell is positive under the preregistered primary rule. Each point is the cell-level median commitment lag  $\Delta_{\text{margin}}$  in raw layer units for the primary analysis path, and each horizontal interval is a grouped-bootstrap 95% confidence interval. All evaluated ARC-Challenge and CommonsenseQA cells are positive. The MMLU abstract-algebra slice contributes several insufficient n cells because the slice itself contains only 11 questions and the preregistration requires at least ten valid examples per model-dataset cell.

| Dataset               | Positive | Evaluable | Insufficient |
|-----------------------|----------|-----------|--------------|
| ARC-Challenge         | 9        | 9         | 0            |
| CommonsenseQA         | 9        | 9         | 0            |
| MMLU Abstract Algebra | 3        | 3         | 6            |
| All datasets          | 21       | 21        | 6            |

Table 2: Primary-rule summary by dataset. ARC-Challenge and CommonsenseQA are positive in every evaluable model-dataset cell. The only non-positive outcomes are insufficient n cells from the small MMLU abstract algebra slice.

## 127 4 Controls and robustness

128 The controls change only one thing: the exact string scored after "Final answer: ". Figure 3  
 129 shows why that matters. Under identity ordering, the primary exact-option readout yields  
 130 a 81% valid-rate for  $\Delta_{\text{margin}}$ . If we keep the prompt fixed but score the literal continuation  
 131 "The answer is {option}" instead, that valid-rate falls to 58%. If we score only the answer  
 132 letter, it rises to 95%.

133 The readouts also disagree on which answer the model prefers at the end. In fully paired  
 134 identity cases, the templated control agrees with the exact-option path on only 60% of cases,  
 135 the letter-label control agrees on only 44%, and all three readouts agree on just 31% of cases.  
 136 This does not weaken the main result. It tells us something important about measurement:  
 137 scoring "A" is not the same experiment as scoring "photosynthesis". The controls are useful  
 138 because they reveal answer-format effects. They are not suitable replacements for the main  
 139 semantic readout.

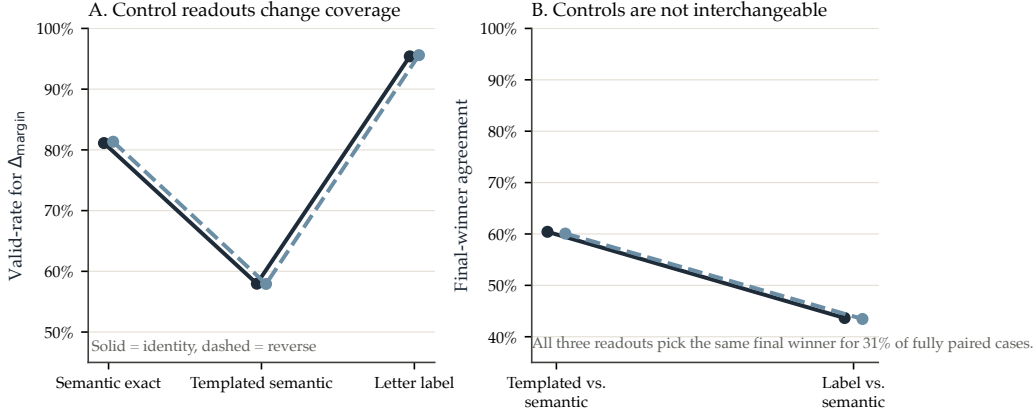


Figure 3: Changing only the answer format changes the measurement. Panel A shows how often  $\Delta_{\text{margin}}$  is defined for each readout. Panel B shows how often the two control readouts pick the same final winner as the semantic-exact primary readout on fully paired cases. Identity and reverse order behave similarly, which is good. The answer-format controls do not, which is exactly why they must stay controls rather than replacing the main semantic path.

140 The answer-order control tells a different story. In Figure 3, the identity and reverse traces  
 141 are nearly on top of each other. That is reassuring because it suggests the main effect is not  
 142 coming from one arbitrary option order. The report-only robustness summaries also line up  
 143 with the primary rule. The secondary criterion yields the same positive-versus-insufficient  
 144 pattern as the primary rule (21 positive, 6 insufficient), and the gold-answer sensitivity  
 145 analysis is positive on all 18 evaluable ARC-Challenge and CommonsenseQA cells while  
 146 remaining underpowered on the MMLU slice (9 insufficient cells).

147 Finally, the QC path is designed to surface problems, not hide them. The released audit  
 148 bundle logs 1224 tokenization collisions and 17 duplicate-option examples instead of silently  
 149 dropping them, while the merged run itself still has zero failure rows and zero unpaired rows.  
 150 The point is not to make the result look tidy. The point is to show that the result still holds  
 151 when we keep the pipeline honest about edge cases.

## 152 5 Interpretation and limits

153 Figure 4 shows what the aggregate result looks like in individual cases. Each panel plots the  
 154 winner-minus-runner contrast for one semantic-exact identity example. The details differ  
 155 by family and prompt, but the same broad pattern appears again and again: the eventual  
 156 answer becomes plausible early, while stable commitment arrives much later and can be  
 157 preceded by hesitation or reversals.

158 These measurements still need to be interpreted carefully. We are not claiming that the  
 159 model literally stores a human-like list of hypotheses, and we are not claiming that these  
 160 curves identify a full causal circuit. Our claim is narrower. Under this prompting and  
 161 scoring setup, the eventual answer often enters the plausible-answer set before it becomes  
 162 stably isolated from the runner-up.

163 The empirical scope is also narrow in three practical ways. First, the paper uses a 1,000-  
 164 example slice rather than the full benchmark splits. Second, the panel contains only  
 165 instruction-tuned open-weight models, so the claim should not be generalized to base models  
 166 or closed systems. Third, MMLU abstract algebra contributes only 11 questions because  
 167 that entire validation slice is tiny in this setup. That is why most of its model-dataset cells  
 168 are underpowered under the preregistered minimum-valid-example rule.

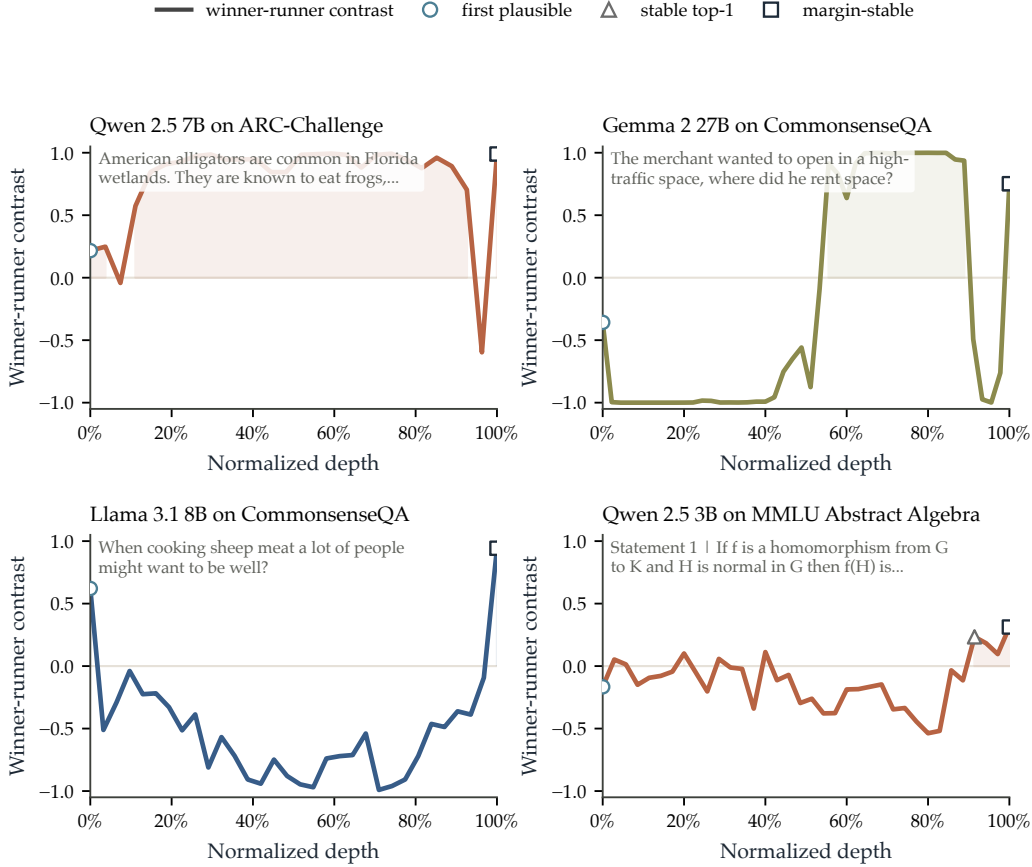


Figure 4: Prompt-level trajectories show the same pattern as the pooled figures. Each panel plots the winner-minus-runner contrast across normalized depth for one semantic-exact identity example. The circular marker shows when the eventual answer first enters the plausible set, the triangular marker shows when it becomes stable top-1, and the square marker shows when it becomes margin-stably committed. The exact shapes vary, but the same left-to-right separation between plausibility and commitment appears at the single-case level.

Those limits matter, but they do not erase the signal. The early-plausibility/late-commitment pattern appears in the pooled depth curves, in every evaluable ARC-Challenge and CommonsenseQA cell, and in the prompt-level trajectories. The controls matter here too. They show that small changes in answer format can change coverage and final winners, which is exactly why the paper keeps exact option-text scoring as the main path.

## 6 Conclusion

The paper’s claim is simple and deliberately scoped. In the setting we study, instruction-tuned models often know which answers are still possible before they have fully settled on the one they will finally say. We show that pattern by ending each prompt with "Final answer: " and tracking how the scores of exact answer continuations change across model depth.

The evidence is consistent across levels of analysis. The flagship pooled curves show a large gap between plausibility and commitment. Every evaluable ARC-Challenge and CommonsenseQA model-dataset cell is positive under the preregistered primary rule. The prompt-level examples show the same separation in concrete trajectories. The answer-format con-

184 trols matter because they show that this result is not automatic: changing the continuation  
185 string can change both coverage and winners.

186 The result does not yet explain why the gap appears, and it should not be overgeneralized  
187 beyond this panel and benchmark slice. But it does support a useful empirical distinction  
188 for future work. The moment when a model finally says an answer can come much later  
189 than the moment when that answer first becomes one of the answers the model already  
190 treats as plausible.

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## 210 A Supplementary material

### 211 A.1 Reproducibility statement

212 This manuscript is generated from a single merged run, pa-  
 213 per\_merged\_9x1000\_20260330T122258Z, with run-scoped manifests, QC summaries,  
 214 bootstrap outputs, and figure/table provenance checked into the paper workspace. All  
 215 numeric claims in the paper are generated from those artifacts rather than entered by hand.  
 216 The merged run contains 1000 prepared prompts, 54,000 scored units, 54,000 metric rows,  
 217 and 258,876 token audits.

### 218 Ethics statement

219 This study analyzes public benchmark questions using open-weight language models and  
 220 does not involve human subjects, personal data, or deployment to end users. The main risk  
 221 is analytical overclaiming: layerwise statistics can be mistaken for a complete mechanistic  
 222 explanation when they are only an operational measurement. We mitigate that risk by  
 223 keeping the claim narrow, preregistering the primary rule, preserving anomaly logs, and  
 224 explicitly reporting the underpowered status of the MMLU slice.

### 225 AI-use disclosure

226 AI-use disclosure. The authors used AI-assisted tools during the project lifecycle: ChatGPT  
 227 for brainstorming and experimental design iteration, Codex for implementation assistance,  
 228 and Claude Opus for drafting and formatting portions of the manuscript and LaTeX source.  
 229 All scientific decisions, experiments, analyses, code review, result verification, and final  
 230 manuscript editing were performed and approved by the human authors, who take full  
 231 responsibility for the paper’s contents.

### 232 A.2 Model registry

| Family            | Model         | Size | Repository                 | Revision prefix |
|-------------------|---------------|------|----------------------------|-----------------|
| Qwen 2.5 Instruct | Qwen 2.5 1.5B | 1.5b | Qwen2.5-1.5B-Instruct      | 989aa7980e4c    |
| Qwen 2.5 Instruct | Qwen 2.5 3B   | 3b   | Qwen2.5-3B-Instruct        | aa8e72537993    |
| Qwen 2.5 Instruct | Qwen 2.5 7B   | 7b   | Qwen2.5-7B-Instruct        | a09a35458c70    |
| Gemma 2 IT        | Gemma 2 2B    | 2b   | gemma-2-2b-it              | 299a8560bedf    |
| 233 Gemma 2 IT    | Gemma 2 9B    | 9b   | gemma-2-9b-it              | 11c9b309abf7    |
| Gemma 2 IT        | Gemma 2 27B   | 27b  | gemma-2-27b-it             | aaf20e6b9f4c    |
| Llama Instruct    | Llama 3.2 1B  | 1b   | Llama-3.2-1B-Instruct      | 9213176726f5    |
| Llama Instruct    | Llama 3.2 3B  | 3b   | Llama-3.2-3B-Instruct      | 0cb88a4f764b    |
| Llama Instruct    | Llama 3.1 8B  | 8b   | Meta-Llama-3.1-8B-Instruct | 0e9e39f249a1    |

234 Table 3: Model registry used in the paper. Short repository names and 12-character revision prefixes are shown for readability; the full pinned revision hashes are preserved in the released provenance manifest.

235 A.3 Full primary positivity table

| Dataset               | Model         | Valid / total | Median lag | 95% CI   | Primary decision |
|-----------------------|---------------|---------------|------------|----------|------------------|
| ARC-Challenge         | Qwen 2.5 1.5B | 109/195       | 22         | [20, 23] | positive         |
| ARC-Challenge         | Qwen 2.5 3B   | 143/195       | 30         | [29, 31] | positive         |
| ARC-Challenge         | Qwen 2.5 7B   | 163/195       | 23         | [22, 24] | positive         |
| ARC-Challenge         | Gemma 2 2B    | 117/195       | 18         | [17, 20] | positive         |
| ARC-Challenge         | Gemma 2 9B    | 132/195       | 29         | [26, 32] | positive         |
| ARC-Challenge         | Gemma 2 27B   | 135/195       | 29         | [25, 34] | positive         |
| ARC-Challenge         | Llama 3.2 1B  | 79/195        | 12         | [11, 13] | positive         |
| ARC-Challenge         | Llama 3.2 3B  | 104/195       | 21         | [20, 22] | positive         |
| ARC-Challenge         | Llama 3.1 8B  | 112/195       | 25         | [24, 26] | positive         |
| CommonsenseQA         | Qwen 2.5 1.5B | 677/794       | 19         | [18, 20] | positive         |
| CommonsenseQA         | Qwen 2.5 3B   | 706/794       | 26         | [25, 26] | positive         |
| CommonsenseQA         | Qwen 2.5 7B   | 718/794       | 22         | [22, 23] | positive         |
| CommonsenseQA         | Gemma 2 2B    | 693/794       | 16         | [15, 17] | positive         |
| CommonsenseQA         | Gemma 2 9B    | 702/794       | 25         | [22, 28] | positive         |
| CommonsenseQA         | Gemma 2 27B   | 706/794       | 27         | [25, 29] | positive         |
| CommonsenseQA         | Llama 3.2 1B  | 625/794       | 10         | [9, 10]  | positive         |
| CommonsenseQA         | Llama 3.2 3B  | 672/794       | 20         | [19, 21] | positive         |
| CommonsenseQA         | Llama 3.1 8B  | 649/794       | 24         | [23, 25] | positive         |
| MMLU Abstract Algebra | Qwen 2.5 1.5B | 7/11          | –          | –        | insufficient n   |
| MMLU Abstract Algebra | Qwen 2.5 3B   | 10/11         | 32         | [28, 33] | positive         |
| MMLU Abstract Algebra | Qwen 2.5 7B   | 10/11         | 25         | [22, 26] | positive         |
| MMLU Abstract Algebra | Gemma 2 2B    | 7/11          | –          | –        | insufficient n   |
| MMLU Abstract Algebra | Gemma 2 9B    | 10/11         | 36         | [24, 40] | positive         |
| MMLU Abstract Algebra | Gemma 2 27B   | 7/11          | –          | –        | insufficient n   |
| MMLU Abstract Algebra | Llama 3.2 1B  | 4/11          | –          | –        | insufficient n   |
| MMLU Abstract Algebra | Llama 3.2 3B  | 2/11          | –          | –        | insufficient n   |
| MMLU Abstract Algebra | Llama 3.1 8B  | 2/11          | –          | –        | insufficient n   |

Table 4: Full cell-level primary-rule results. The primary decision is evaluated separately for each model-dataset cell on the semantic-exact identity path.

238 A.4 Selection and QC details

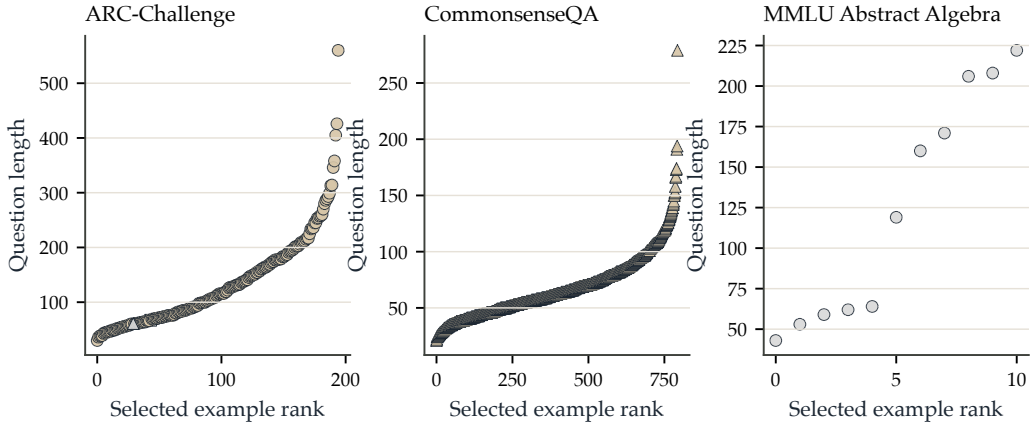


Figure 5: The paper slice spans a wide range of selected prompt lengths. The figure shows the selected-example question-length profile for each dataset together with the chosen option-count markers. MMLU is exhausted because the abstract-algebra validation slice contains only 11 prompts.

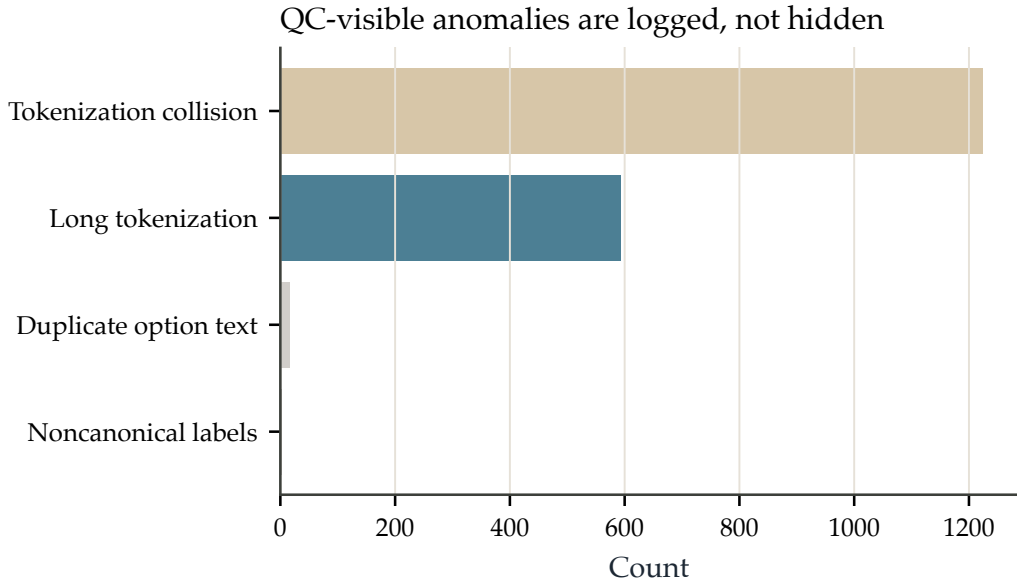


Figure 6: QC-visible anomalies are logged rather than hidden. These counts come directly from the merged run’s QC summary. They motivate caution, but they do not arise from silent exclusion or failed pairing.

| QC item                                | Count  |
|----------------------------------------|--------|
| Prepared examples                      | 1000   |
| Token audits                           | 258876 |
| Tokenization collisions                | 1224   |
| Suspiciously long option tokenizations | 594    |
| Duplicate option text                  | 17     |
| Noncanonical option labels             | 2      |
| Unpaired rows                          | 0      |
| Failure rows                           | 0      |

Table 5: QC summary for the merged run. The run completes with zero failure rows and zero unpaired rows while still surfacing tokenization and option-surface anomalies.

| Readout            | Continuation surface                                           | Normalization        |
|--------------------|----------------------------------------------------------------|----------------------|
| semantic_exact     | Exact option text after "Final answer:", e.g. "photosynthesis" | Mean log-probability |
| templated_semantic | Literal continuation "The answer is {option_text}"             | Mean log-probability |
| letter_label       | Bare answer letter after "Final answer:", e.g. "A"             | Mean log-probability |

Table 6: Readout definitions used in the paper. The controls intentionally differ in continuation surface while keeping the rest of the setup fixed.